

Category Table

```
CREATE TABLE CATEGORY (  
    CategoryID INT PRIMARY KEY AUTO_INCREMENT,  
    CategoryName VARCHAR(100) NOT NULL  
);
```

Product Table

```
CREATE TABLE PRODUCT (  
    ProductID INT PRIMARY KEY AUTO_INCREMENT,  
    ProductName VARCHAR(100) NOT NULL,  
    CategoryID INT NOT NULL,  
    Price DECIMAL(10,2) NOT NULL,  
    Stock INT NOT NULL,  
    FOREIGN KEY (CategoryID) REFERENCES Category(CategoryID)  
);
```

Table Structure

Category

Column Name	Data Type	Constraint
CategoryID	INT	PRIMARY KEY, AUTO_INCREMENT
CategoryName	VARCHAR(100) NOT NULL	

Product

Column Name	Data Type	Constraint
ProductID	INT	PRIMARY KEY, AUTO_INCREMENT
ProductName	VARCHAR(100)	NOT NULL
CategoryID	INT	FOREIGN KEY, NOT NULL
Price	DECIMAL(10,2)	NOT NULL
Stock	INT	NOT NULL

This design establishes a **one-to-many relationship**: one category can contain many products.